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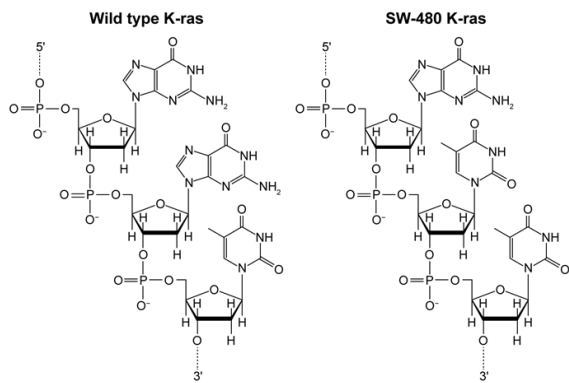


Figure 1 Structural comparison of K-ras codon 12 in wild-type cells and SW-480 cells

PCR sequence analysis has been the preferred method for studying genetic mutations such as those in K-ras. However, challenges in analyzing PCR amplicons have prompted the development of DNA melting curve analysis (MCA). In the MCA technique, DNA is extracted from clinical samples and amplified by PCR. The amplified DNA is then melted (30-70°C) and analyzed by assessing the fluorescence (at 640 nm) of LC Green Plus, an intercalating dye that fluoresces when bound to double-stranded DNA (Figure 2).

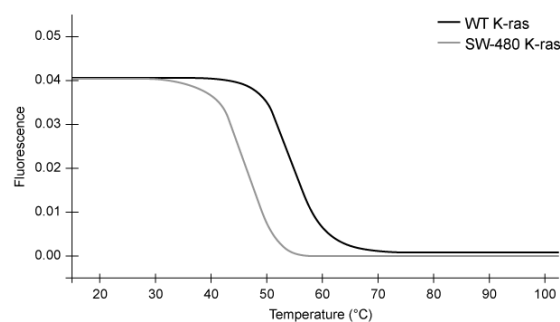


Figure 2 Melting curves of wild-type (WT) and SW-480 K-ras DNA.

To study K-ras activity in colorectal cancer cells, short tandem repeats (STRs) were introduced into the K-ras gene. STRs are short stretches of DNA with repeating nucleotides, and four different STR sequences were inserted into the K-ras gene and analyzed via MCA under different experimental conditions (Table 1).

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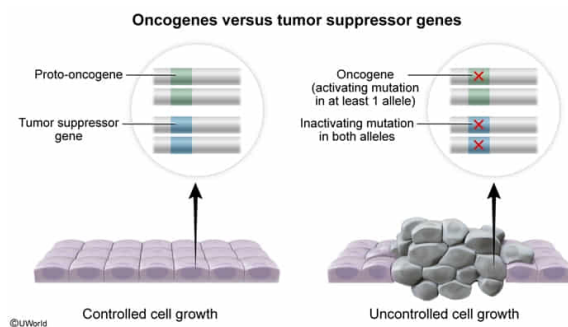
Which of the following is most likely to be identified in SW-480 cells?

- ✓ ☒ A. An activating mutation in at least one K-ras allele [62%]
- ☐ B. Inactivating mutations in both K-ras alleles [8%]
- ☐ C. Decreased expression of K-ras protein [15%]
- ☐ D. Increased expression of hypoactive K-ras protein [14%]

Correct

61% answered correctly

Explanation:



Cancer cells are able to grow and divide excessively due to loss of cell cycle control. Various types of cancers result from **oncogene activation**, **tumor suppressor gene inactivation**, or both.

- **Proto-oncogenes** are wild-type (WT) genes that are normally involved in cell cycle progression (eg, cell division stimulation, apoptosis prevention). However, when proto-oncogenes are mutated into **oncogenes** via **activating mutations** in one or two alleles, cells become cancerous. As a result, oncogenes lead to either an overexpressed protein, a hyperactive protein, or both, promoting uncontrolled cell proliferation.
- **Tumor suppressor genes** (or antioncogenes) regulate DNA repair by repressing or pausing the cell cycle to ensure that only normal cells proceed to the division (mitosis) stage. These genes also prevent accumulation of mutations in cancer cells by either repairing the mutations or inducing programmed cell death (apoptosis) if repair fails. In the setting of cancer, tumor suppressor genes become inactivated by *loss of function mutations in both alleles* and lose the ability to prevent abnormal growth and division of damaged cells.

According to the passage, the K-ras gene normally encodes a protein responsible for activating the signaling pathways that promote DNA replication (cell cycle S-phase) along with cell growth and survival. Mutated K-ras in SW-480 cells is most likely constantly inducing signaling pathways responsible for the characteristic uncontrolled growth, division, and survival of cancer cells. Therefore, **K-ras** is most likely a **proto-oncogene** with an **activating mutation** in at least one allele that turned it into an **oncogene**.

(Choice B) Inactivating mutations in tumor suppressors can lead to cancer, but K-ras is a

proto-oncogene.

(Choices C and D) In cancer, the mutated K-ras protein has either normal function with increased expression or hyperactive (not hypoactive) function with normal expression.

Educational objective:

Oncogenes are mutated forms of genes involved in cell cycle progression (proto-oncogenes) that transform normal cells into cancer cells. Tumor suppressor genes regulate, pause, and inhibit cycle cell progression to ensure damaged DNA is repaired. In cancerous cells, mutations inactivate tumor suppressor genes and activate oncogenes.

Time spent:0 sec

Subject:
Biochemistry

QID:401983

System:
Carbohydrates, Nucleotides, and Lipids

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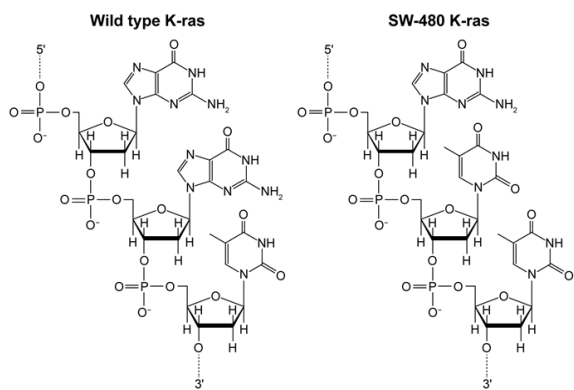


Figure 1 Structural comparison of *K*-ras codon 12 in wild-type cells and SW-480 cells

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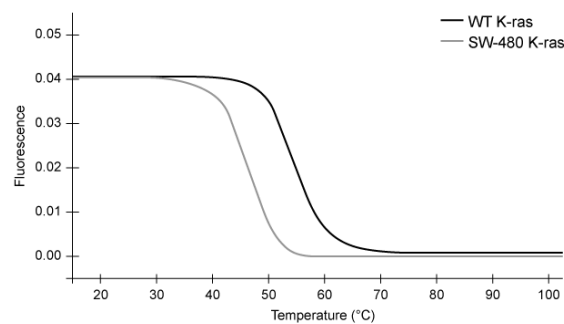


Figure 2 Melting curves of wild-type (WT) and SW-480 K-ras DNA.

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What are the physical states of wild-type (WT) and SW-480 K-ras DNA at 55°C?

- ☐ A. 100% of SW-480 DNA is double-stranded and 50% of WT DNA is single-stranded. [4%]
- ✓ ☒ B. 100% of SW-480 DNA is single-stranded and 50% of WT DNA is single-stranded. [74%]
- ☐ C. 50% of SW-480 DNA is double-stranded and 100% of WT DNA is double-stranded. [11%]
- ☐ D. 50% of SW-480 DNA is single-stranded and 100% of WT DNA is double-stranded. [9%]

Correct

73% answered correctly

Explanation:

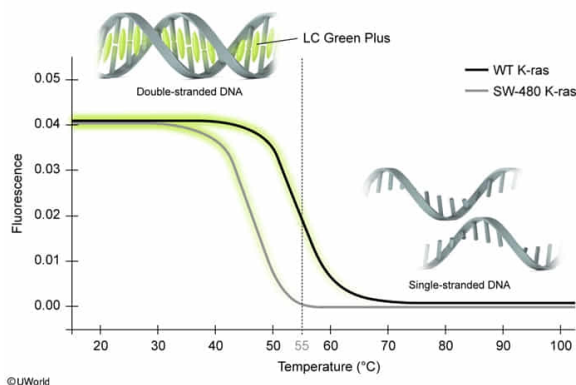


Figure 2 shows the DNA melting curve for wild-type (WT) K-ras DNA and mutant K-ras DNA produced by SW-480 cells. To generate a melting curve, identical copies of each type of double-stranded DNA are gradually exposed to increasing temperatures, which causes the separation (denaturation) of the double helix into two strands of DNA. The **melting temperature (T_m)**, depicted as the **midpoint of the melting curve**, is achieved when **50% of double-stranded DNA has become single-stranded** (ie, when half of the helical structure is lost).

LC Green Plus is a compound that fluoresces strongly when bound to double-stranded DNA but has minimal fluorescence with single-stranded DNA. The amount of double-stranded DNA in a sample can be detected by measuring the fluorescence of LC Green Plus. Consequently, as temperature increases in DNA melting curves, the fluorescence decreases as double-stranded DNA becomes single-stranded.

In Figure 2, both WT K-ras and SW-480 K-ras DNA have a maximal fluorescence of 0.04

at 20°C, which signals that all DNA molecules are double-stranded and bound by LC Green Plus. The fluorescence of WT K-ras DNA decreases to 0.02 **at 55°C**, which is the T_m of WT K-ras DNA and signifies that **50% of the WT DNA has melted and become single-stranded** while the other 50% is double-stranded. However, mutant K-ras DNA from SW-480 cells reacts differently at 55°C; the fluorescence decreases to a value near zero, indicating that **100% of mutant K-ras DNA is single-stranded**.

(Choices A, C, and D) At 55°C, Figure 2 shows that 100% of SW-480 K-ras DNA is single-stranded (*not* double-stranded) but that 50% (*not* 100%) of WT DNA is single-stranded.

Educational objective:

To produce a melting curve, double-stranded DNA is gradually exposed to increasing temperatures, which cause the separation of the double helix into two strands of DNA. The melting temperature (T_m) is achieved when 50% of double-stranded DNA has become single-stranded; T_m is usually depicted as the midpoint of the melting curve.

Time spent:0 sec

Subject:
Biochemistry

QID:401984

System:
Carbohydrates, Nucleotides, and Lipids

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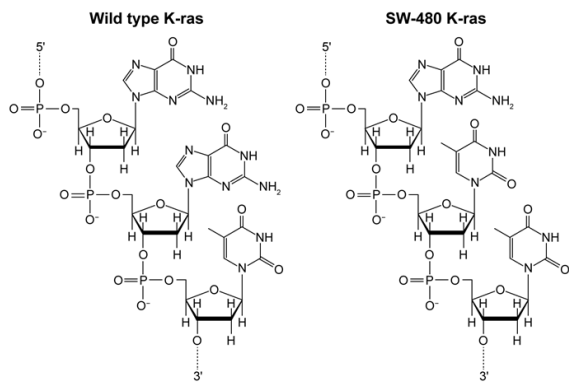


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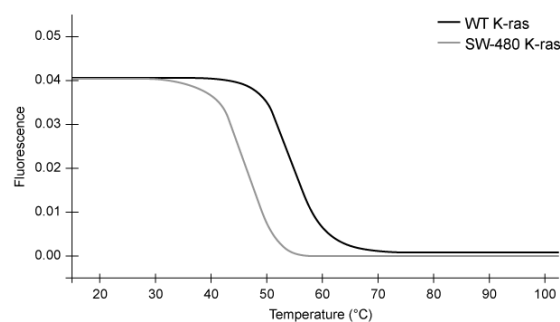


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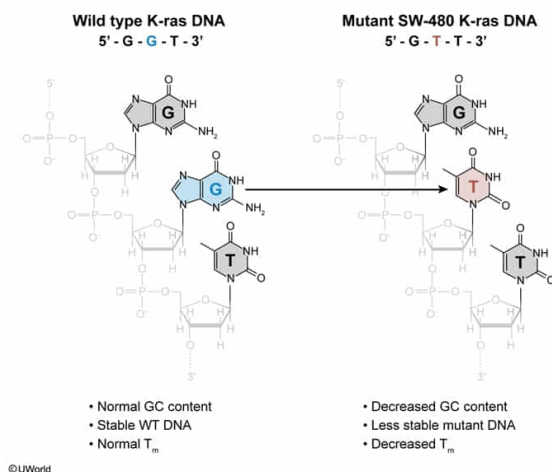
Assuming that additional mutations similar to the one shown in Figure 1 occur throughout the mutant K-ras gene, what would be the best explanation for the difference in denaturation between wild-type (WT) and mutant K-ras DNA?

- ☐ A. Mutant DNA has a higher GC content than WT DNA. [11%]
- ✓ ☒ B. Mutant DNA has a lower GC content than WT DNA. [85%]
- ☐ C. WT DNA is longer than mutant DNA. [2%]
- ☐ D. WT DNA is shorter than mutant DNA. [1%]

Correct

84% answered correctly

Explanation:



The GC content is the percentage of guanine and cytosine bases in DNA, and **the higher the GC content, the greater the stability and melting temperature (T_m) of double-stranded DNA**. DNA double helix stability is primarily maintained by stacking interactions: the interactions between neighboring $G\equiv C$ base pairs are stronger than those between adjacent $A=T$ base pairs. It also takes more energy (heat) to break the three hydrogen bonds in $G\equiv C$ base pairs than the two hydrogen bonds in $A=T$ base pairs. As a result, the T_m for GC-rich DNA molecules is higher as additional energy is required to separate the more stable helix.

Figure 1 shows that SW-480 cells have a $G\rightarrow T$ base substitution at codon 12 of the K-ras gene, depicting a loss in GC content. Assuming there are additional similar mutations in the mutant K-ras gene, the **mutant DNA will have lower GC content** than the WT DNA and therefore be less stable and denature faster (ie, **exhibit lower T_m**).

(Choice A) Mutant K-ras DNA with various mutations similar to the one shown in Figure 1 would exhibit a loss of GC content and decreased stability and T_m .

(Choices C and D) WT DNA and mutant K-ras DNA are the same length because the mutation in K-ras from SW-480 cells is a base substitution and not an insertion or

deletion. Longer DNA molecules have increased T_m because they have more base pairs that stabilize the double helix and require more energy to be broken.

Educational objective:

G≡C base pairs form more hydrogen bonds and have stronger stacking interactions than A=T base pairs. The higher the GC content, the greater the melting temperature (T_m) and stability of double-stranded DNA.

Time spent:0 sec

Subject:
Biochemistry

QID:401985

System:
Carbohydrates, Nucleotides, and Lipids

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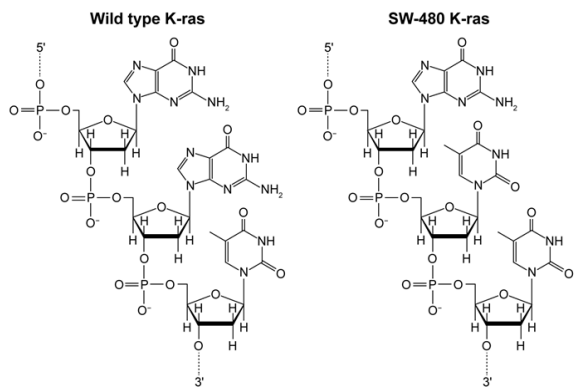


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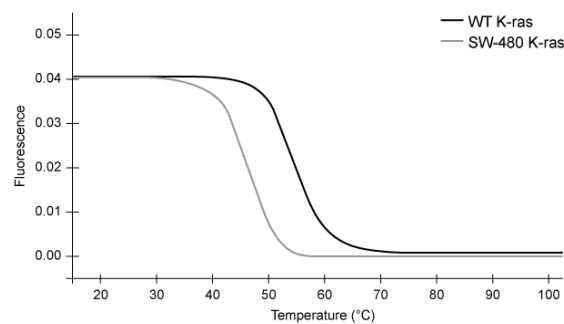


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Prior to inserting the STR sequences into K-ras DNA, the researchers perform MCA analysis using only the double-stranded STR sequences. Which sequence from Table 1 would have the fastest reannealing time in this experiment?

- ☐ A. Sequence 1 [4%]
- ☐ B. Sequence 2 [27%]
- ✓ ☒ C. Sequence 3 [61%]
- ☐ D. Sequence 4 [5%]

Correct

62% answered correctly

Explanation:

DNA annealing speed depends on length, pH, and salt content

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Fastest →
annealing

Shortest strand ~physiological pH Highest salt concentration

Melted DNA strands will **reanneal** to form the double helix once temperature decreases, but the time it takes for complementary strands to reanneal depends on the following:

- **DNA length:** Longer DNA molecules have more hydrogen bonds and will take more time to both melt and reanneal.
- **pH:** The physiological pH range (7.3-7.4) allows maximal hydrogen bonding between nitrogenous bases of DNA. At low pH, hydrogen bond acceptor atoms in the bases become protonated, and these protonated acceptors cannot form hydrogen bonds, causing double helix separation. In contrast, high pH causes deprotonation of hydrogen bond donors in the bases; the loss of protons results in the loss of hydrogen bonds and a destabilized double helix.
- **Salt concentration (ionic strength):** The electrostatic repulsion between negatively charged phosphate groups on the sugar-phosphate backbone destabilizes the double helix. However, this repulsion is neutralized and shielded by the binding of positively charged species in solution (eg, Na^+ and Mg^{2+} cations). High salt concentration of the solution increases double helix stability, but low salt concentration decreases stability.

The four short tandem repeats sequences shown in Table 1 were denatured during melting curve analysis and cooled in environments with different pH and NaCl concentrations. **Sequence 3** will have the **fastest reannealing time** as it is the **shortest**

sequence and was **allowed to cool** in the **solution closest to physiological pH with a high salt concentration** ($>0.15\text{ M}$).

(Choices A and B) Sequences 1 and 2 were studied in solutions with high pH and low salt concentration. Alkaline pH causes a loss of base pairing interactions, and low salt concentration would induce greater repulsion between negatively charged phosphate groups on DNA. In addition, sequence 2 is the longest and would take longer to reanneal.

(Choice D) Sequence 4 DNA has a higher number of repeats than sequence 3 and is likely to be destabilized by the extremely acidic pH despite the high salt concentration.

Educational objective:

The following parameters influence the thermodynamic stability of the DNA duplex:

- **DNA length:** Longer DNA molecules take more time to both melt and reanneal.
- **pH:** Extreme changes in pH outside the physiological range lead to loss of hydrogen bonding and destabilize the DNA helix.
- **Salt concentration (ionic strength):** High salt concentration of the solution increases double helix stability, but decreased salt concentration decreases stability.

Time spent:0 sec

Subject:
Biochemistry

QID:401986

System:
Carbohydrates, Nucleotides, and Lipids

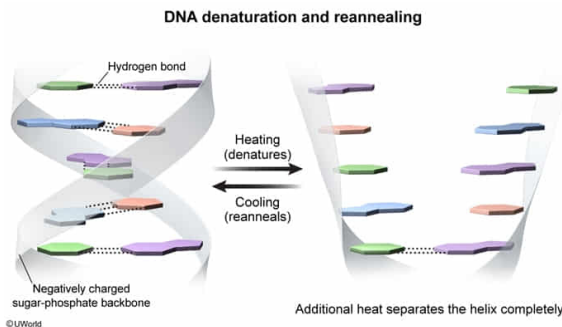
Which factor decreases the stability of a DNA double helix during denaturation?

- ☐ A. Base stacking is enhanced.
- ☐ B. Hydrophobic effects are increased. [1%]
- ✓ ☒ C. Hydrogen bonds are disrupted. [87%]
- ☐ D. Phosphodiester bonds are broken. [10%]

Correct

87% answered correctly

Explanation:



The DNA double helix is arranged such that the sugar-phosphate backbone (linked by phosphodiester bonds) lies on the outside and the nitrogen-containing bases (paired by hydrogen bonds) lie on the inside. During double helix formation, complementary strands of DNA anneal through the creation of hydrogen bonds between the nitrogenous bases. Once the double helix is formed, energy is required to denature the duplex into two separate strands. **Increasing the temperature** during the melting step provides the **heat energy** necessary to **break the hydrogen bonds** between bases and separate the double helix.

(Choice A) The base pairs in the double helix are "stacked" almost on top of each other; however, base stacking is *interrupted*, not enhanced, when the DNA helix denatures during melting. Stacking interactions, which are due to hydrophobic effects and nonpolar van der Waals forces between the bases, are primarily responsible for the stability of the double helix structure.

(Choice B) Hydrophobic effects decrease during the melting step because hydrophobic bases are exposed to water. In aqueous solution, hydrophobic effects force hydrophobic bases toward the double helix interior and away from the polar solvent (water). In contrast, the hydrophilic, negatively charged phosphates in the backbone can hydrogen bond with water molecules.

(Choice D) Covalent phosphodiester bonds of the sugar-phosphate backbone maintain the structure of single-stranded DNA. These linkages are not affected when the double helix is separated during denaturation.

Educational objective:

During DNA double helix formation, hydrogen bonds form between the bases and release energy. Heat energy provided in a DNA melting reaction is used to disrupt interactions that contribute to DNA double helix structure: hydrogen bonding, base stacking, and hydrophobic effects.

Time spent:0 sec**Subject:**

Biochemistry

QID:401987**System:**

Carbohydrates, Nucleotides, and Lipids

The double-stranded coding region of a certain gene consists of a total of 564 nucleotides, 135 of which are adenosine residues. If the sense strand contains 75 adenosine residues, then all of the following must be true EXCEPT:

- ☐ A. The antisense strand contains 75 thymidine residues. [6%]
- ☐ B. The sum of the guanosine and cytidine residues in the sense strand is 147. [28%]
- ✓ ☒ C. Half of the base pairs in the coding region are adenosine and thymidine residues. [47%]
- ☐ D. The sum of the adenosine and guanosine residues in the coding region equals the sum of the thymidine and cytidine residues. [16%]

Correct

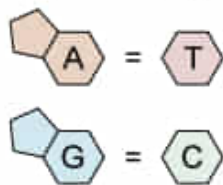
48% answered correctly

Explanation:

Chargaff's rules

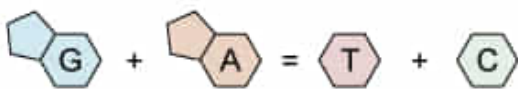
Rule 1

Purines = Pyrimidines



Rule 2

Sum of purines = Sum of pyrimidines



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Purine nucleoside residues (guanosine or adenosine) **base pair** with pyrimidine nucleoside residues (cytidine or thymidine) in DNA. Adenosine (A) base pairs exclusively with thymidine (T) via two hydrogen bonds, and guanosine (G) base pairs with cytidine (C) via three hydrogen bonds. As a result, **Chargaff's rules** state that for any double-stranded DNA molecule:

- The number of guanosine equals the number of cytidine, and the number of adenosine equals the number of thymidine (ie, **G = C** and **A = T**).
- The sum of the purines equals the sum of the pyrimidines (ie, **G + A = C + T**).

The question states that the double-stranded coding region of a gene contains 564 total nucleotides, and that 135 of these are adenosine; it then asks for the statement that is *not* true. Each of the following statements *are* true, and can be ruled out:

- The question states that 75 of the 135 adenosine residues are found in the sense strand. Each of these residues must base pair with a thymidine residue in the antisense strand. Therefore, the **antisense strand contains 75 thymidine residues (Choice A)**.
- The coding region contains 564 nucleotides, of which 135 are adenosine residues, and another 135 must be thymidine residues. Therefore, the remaining 294 residues must be guanosine and cytidine residues ($564 - 135 - 135 = 294$). Half of these residues ($294/2 = 147$) are in the sense strand and half are in the antisense strand, so **the sense strand contains 147 total guanosine and cytidine residues (Choice B)**.
- According to Chargaff's rule, the sum of the purines (adenosine and guanine) in any double-stranded DNA molecule equals the sum of the pyrimidines (thymidine and cytidine) (**Choice D**).

Because the coding region contains 135 adenosine residues, 135 of the base pairs must be between adenosine and thymidine residues. There are a total of 282 base pairs in the coding region ($564/2 = 282$), half of which is 141. Therefore the 135 A-T base pairs make up *less* than half of the base pairs in the coding region (ie, <50% A-T content, >50% G-C content).

Educational objective:

Chargaff's rules state that for any double-stranded DNA sample, a 1:1 nucleotide ratio of pyrimidines to purines must exist, and they are arranged such that:

- $A = T$ and $G = C$
- $G + A = C + T$

Time spent: 0 sec
Subject:
Biochemistry

QID: 401988
System:
Carbohydrates, Nucleotides, and Lipids